



What's past is past: Neither perceptual preactivation nor prior motivational relevance decrease subsequent inattentional blindness

Carina Kreitz^{a,*}, Robert Schnuerch^{b,1}, Philip A. Furley^{a,2}, Daniel Memmert^{a,2}

^a Institute of Training Science and Sport Informatics, German Sport University Cologne, Cologne, Germany

^b Department of Psychology, University of Bonn, Bonn, Germany

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ABSTRACT

Inattentional blindness—the phenomenon that clearly visible, yet currently unexpected objects go unnoticed when our attention is focused elsewhere—is an ecologically valid failure of awareness. It is currently subject to debate whether previous events and experiences determine whether or not inattentional blindness occurs. Using a simple two-phase paradigm in the present study, we found that the likelihood of missing an unexpected object due to inattention did not change when its defining characteristic (its color) was perceptually preactivated (Experiment 1; $N = 188$). Likewise, noticing rates were not significantly reduced if the object's color was previously motivationally relevant during an unrelated detection task (Experiment 2; $N = 184$). These results corroborate and extend recent findings questioning the influence of previous experience on subsequent inattentional blindness. This has implications for possible countermeasures intended to thwart the potentially harmful effects of inattention.

1. General introduction

Human conscious perception is severely limited, which becomes evident in a variety of failures of awareness (Dehaene, Changeux, Naccache, Sackur, & Sergent, 2006). Perhaps the most striking and ecologically valid example is *inattentional blindness*, the phenomenon that even salient unexpected objects go unnoticed when attention is focused elsewhere (Mack & Rock, 1998; Simons & Chabris, 1999). The consequences of these misses in real-world settings are potentially severe, such as in road traffic (Most & Astur, 2007), aviation (Haines, 1991), or medical diagnostics (Drew, Vo, & Wolfe, 2013). Hence, it is highly expedient to understand what affects the likelihood of inattentional blindness (e.g., Downing, Bray, Rogers, & Childs, 2004; Newby & Rock, 1998). Several studies suggest that it depends on *stable features of the observer*, such as cognitive capabilities (Hannon & Richards, 2010; O'Shea & Fieo, 2015) and personality traits (Richards, Hellgren, & French, 2014). However, this view has been contested in several recent publications (e.g., Bredemeier & Simons, 2012; Kreitz, Furley, Memmert, & Simons, 2015; Kreitz, Schnuerch, Gibbons, & Memmert, 2015). The notion that *situational parameters*, such as attributes of the unexpected object, have an effect on the likelihood of inattentional blindness is much less controversial (see, e.g., Calvillo & Jackson, 2014; Mack & Rock, 1998). Perhaps most importantly,

* Corresponding author at: Institute of Training Science and Sport Informatics, German Sport University Cologne, Am Sportpark Müngersdorf 6, 50933 Cologne, Germany.

E-mail addresses: c.kreitz@dshs-koeln.de (C. Kreitz), robert.schnuerch@uni-bonn.de (R. Schnuerch), p.furley@dshs-koeln.de (P.A. Furley), d.memmert@dshs-koeln.de (D. Memmert).

¹ Department of Psychology, University of Bonn, Kaiser-Karl-Ring 9, 53111 Bonn, Germany.

² Institute of Training Science and Sport Informatics, German Sport University Cologne, Am Sportpark Müngersdorf 6, 50933 Cologne, Germany.

the detection of unexpected objects depends on a person's current attentional set: Unexpected objects are substantially more likely to be perceived when they are similar to the objects that are being attended to as part of the primary task a person is engaged in, for example, regarding perceptual features such as luminance, shape, or color (Koivisto & Revonsuo, 2008; Most, Scholl, Clifford, & Simons, 2005; Most et al., 2001).

While the role of *current* situational features of inattention blindness tasks has been thoroughly investigated, relatively little is known about the effects of *previous* events on subsequent inattention blindness. It seems reasonable to assume that one determinant of whether or not we notice something unexpected might be experience from our immediate past. After all, there is little doubt that perception is not only driven by concurrent stimulation, but also substantially shaped by previous events (see, e.g., Olivers, Meijer, & Theeuwes, 2006; Soto, Heinke, Humphreys, & Blanco, 2005; Stokes, Atherton, Patai, & Nobre, 2012). Indeed, there is some evidence that prior experience—broadly construed—does affect the likelihood of inattention blindness. *First*, indirectly alerting somebody to the possibility that additional events can occur influences whether or not they detect these events in the immediate future (Most et al., 2001): Participants who have recently been queried about a potentially overlooked unexpected object are subsequently less likely to miss another spontaneously appearing object (e.g., Kreitz, Furley, et al., 2015; Most et al., 2001). *Second*, Rattan and Eberhardt (2010) suggested that simply preactivating a certain social concept subsequently reduces the likelihood of missing an unexpected stimulus erroneously associated with that concept. *Third*, subliminally priming a detection goal decreases inattention blindness rates significantly (Légal, Chekroun, Coiffard, & Gabarrot, 2017).

However, the notion that prior experience affects inattention blindness is not undisputed. *First*, the abovementioned effect of previous knowledge is relatively limited. An abstract, generalized knowledge of the phenomenon or of the possibility that additional events can occur does not suffice to reduce the likelihood of missing spontaneously appearing stimuli. Rather, a relatively specific expectation about potential additional objects in the given context must be formed to diminish inattention blindness (Beanland & Pammer, 2010; Kreitz, Furley, et al., 2015; Ward & Scholl, 2015). *Second*, we recently failed to observe clear-cut effects of preactivation. In two experiments (with a total of 221 participants), we demonstrated that participants did not detect an unexpected colored shape more frequently after previous engagement with the concept of this very color (Kreitz, Schnuerch, Furley, Gibbons, & Memmert, 2015): Neither mentioning the color several times before the appearance of the unexpected stimulus nor priming the color implicitly via related concepts significantly reduced the susceptibility to inattention blindness. Fittingly, detection rates of an unexpectedly appearing gorilla were not affected by subliminally priming the semantic concept of a gorilla (Légal et al., 2017).

As the evidence regarding the potential influence of prior experience on the likelihood of inattention blindness is somewhat mixed, we further investigated this issue in the present study. We revisited the idea that preexposure to certain stimulations in one context might determine whether or not we subsequently notice related stimuli in another context, even if they are unexpected (Kreitz, Schnuerch, Furley, et al., 2015; Rattan & Eberhardt, 2010). Specifically, we tested whether perceptual preactivation of a color reduces the likelihood of missing an object of this color (Experiment 1) and whether a color that is motivationally relevant for a while is later noticed more frequently (Experiment 2).

2. Experiment 1

As mentioned above, previous work suggests that *semantic* preactivation of a certain color does not affect whether an unexpected object crosses the threshold of awareness in an inattention blindness scenario (Kreitz, Schnuerch, Furley, et al., 2015; Légal et al., 2017; but see Rattan & Eberhardt, 2010). In the first experiment, we addressed an even more basic issue and investigated whether *perceptual* preactivation influences inattention blindness. Prior studies in various fields have shown that perceiving a stimulus earlier in time influences how this stimulus is perceived (and attended to) at a later stage. Priming studies indicate that objects are identified more accurately and more quickly if the object was presented before (Tulving & Schacter, 1990; Wiggs & Martin, 1998). This facilitatory effect of perceptual repetition not only applies to entire objects, but also to individual features, such as colors (e.g., Goolsby & Suzuki, 2001). Based on these findings, it seems feasible that even the threshold of declarative awareness of an unexpected object might be crossed due to prior activation of the object's central features. It is widely assumed that sensory input needs to pass a certain threshold to reach awareness, which is often achieved by additional amplification (Dehaene et al., 2006). Perceptual preactivation might be an important means for such amplification, allowing otherwise unnoticed stimuli to be consciously detected (Kreitz, Schnuerch, Furley, et al., 2015).

To test this idea, we conducted a straightforward experiment: Participants first performed a simple task in which they were (or were not) repeatedly and incidentally exposed to a certain color (priming phase). In a second task, an object of the same or a different color appeared unexpectedly (inattention blindness phase). We assessed whether the unexpected object was detected more frequently if its color had previously been presented for several minutes (i.e., perceptually preactivated) than if it had not been presented.

2.1. Method

2.1.1. Participants

A total of 200 participants took part in Experiment 1. All participants gave written informed consent and received candy as compensation. We excluded participants from analysis if they (a) indicated in the follow-up questionnaire that they had anticipated the unexpected object or knew that inattention blindness was the subject of the study (7 participants) or (b) did not notice the unexpected object in the control condition in which they did not have to perform the primary task (full-attention trial; 4 participants). All participants reported normal or corrected-to-normal vision and performed adequately during the priming phase (no outliers).

regarding errors; all participants within 2 *SD* of the overall sample mean). Thus, no one had to be excluded from further analysis based on these criteria. Data of one participant were incomplete due to an error in recording. Data from the remaining 188 participants were analyzed ($M = 23.3$ years; $SD = 5.8$ years; 47.3% female).

2.1.2. Materials and procedure

Participants signed a declaration of consent and were seated at a distance of approximately 50 cm from a 24-in. computer display (resolution: 1920×1080 pixels). Participants first completed the priming phase and then—without delay—the inattentional blindness phase. Both tasks were programmed and run on Presentation (Neurobehavioral Systems, Berkeley, CA) and participants responded using a standard keyboard. They were tested alone or in pairs (dividers separated the two work spaces). Instructions were given on screen prior to each task. Participants were randomly assigned to one of the two experimental conditions: The color of the unexpected object in the inattentional blindness phase was either preactivated in the priming phase (preactivation condition) or was not preactivated (control condition). Also, half of the participants were presented with a blue unexpected object and half of the participants were presented with an orange unexpected object. After completion of both the priming phase and the inattentional blindness phase, participants filled out a questionnaire collecting demographics, general knowledge about inattentional blindness, and anticipation of the unexpected object. Finally, participants were debriefed.

2.1.2.1. Priming phase. In the priming phase, participants' task was to fixate the center of the screen and count the number of target circles (i.e., those framed by a black annulus) in each block. A total of 220 trials was presented in eleven blocks with 20 trials each. After each block, participants entered the number of target circles. Each trial began with a fixation cross that was shown for an average of 800 ms (jittered between 600 and 1000 ms), followed by a circle with a diameter of 2.5° visual angle that was presented for 200 ms. Each circle appeared randomly at one of four possible positions (45° , 135° , 225° , 315°) on an imaginary circle (radius: 3.2° visual angle) surrounding the central fixation cross. Before the next trial commenced, a 1000-ms delay was inserted.

The background during the entire trial was grey (RGB: 128,128,128). In the control condition, no colors appeared during the priming phase: half of the circles were white (RGB: 255,255,255) and half were grey (RGB: 150,150,150). In the preactivation condition, all circles were colored: half of them were orange (RGB: 255,127,0) and half of them were blue (RGB: 0,127,255). In both conditions, half of the circles (random in regard to the circles' color) were framed by a black annulus. Those circles had a total diameter of 2.9° visual angle and were targets that participants were instructed to count in each block. There were 5–15 targets in each of the eleven blocks with no target count being used twice (i.e., the number of targets in each block was different than in all other blocks). Participants silently counted the number of targets for each block and were asked to give the number after each block. They did not receive feedback on their count. An example trial of the priming phase is depicted in Fig. 1.

2.1.2.2. Inattentional blindness phase. The task was adapted from a very similar procedure reported by Mack and Rock (1998): Participants repeatedly judged which of the two arms of a briefly presented cross was longer. Following a critical trial, they were asked whether they had noticed the appearance of an unexpected shape. Each trial began with a 1000-ms fixation screen, followed by a cross for 200 ms and then by a black-and-white pattern mask for 100 ms (see Fig. 1). Participants judged which line of the cross (i.e., the horizontal one or the vertical one) was longer, responding via predefined keys on a standard keyboard. Whether the horizontal or the vertical arm was longer was determined randomly on each trial. Each participant completed five easy practice trials (longer arm 6° , smaller arm 4.5°) followed by five more difficult and attention-demanding trials (longer arm 6° , smaller arm 5.62° ³). The eleventh trial was the critical trial and was presented immediately and without forewarning. It had the same difficulty as trials six to ten, yet included an unexpected square ($0.9^\circ \times 0.9^\circ$) which appeared along with the cross for the entire 200 ms. The square was always presented with a distance of 2° from the center on one of the imaginary 45° lines bisecting the quadrants defined by the cross. The particular quadrant in which the square appeared was chosen randomly for each participant. The color of the square was counterbalanced across participants (RGB orange: 255,127,0; RGB blue: 0,127,255). After manually indicating which line they thought was longer, participants were asked if they had seen anything other than the cross which had not been present before. Irrespective of their answer, participants were then asked in which part of the display the additional object had been presented (forced choice: upper right, lower right, lower left, upper left) and which shape (six choices) it had. They were told to guess if they had not noticed anything. Following the critical trial, there was a final trial in which participants were instructed to still fixate on the center of the display, but they did not have to perform the line-judgment task any more (full-attention trial). The position of the additional square was again chosen randomly while its color was the same as before for each participant. The questions concerning the additional square were exactly the same as those presented before.

2.2. Results and discussion

Participants were coded as inattentionally blind if they did not report noticing the unexpected object or claimed to have seen something but could not define either its location or its shape (see, e.g., Kreitz, Schnuerch, Furley, et al., 2015). Only data from participants who noticed the additional colored square in the control condition (full-attention trial) and correctly identified its position or its shape were analyzed. Thus, missing the unexpected object in the critical trial cannot be attributed to basal visual

³ This value was the average length yielding an accuracy threshold of 79% in participants who completed this exact same task in previous studies (e.g., Kreitz, Furley, et al., 2015).

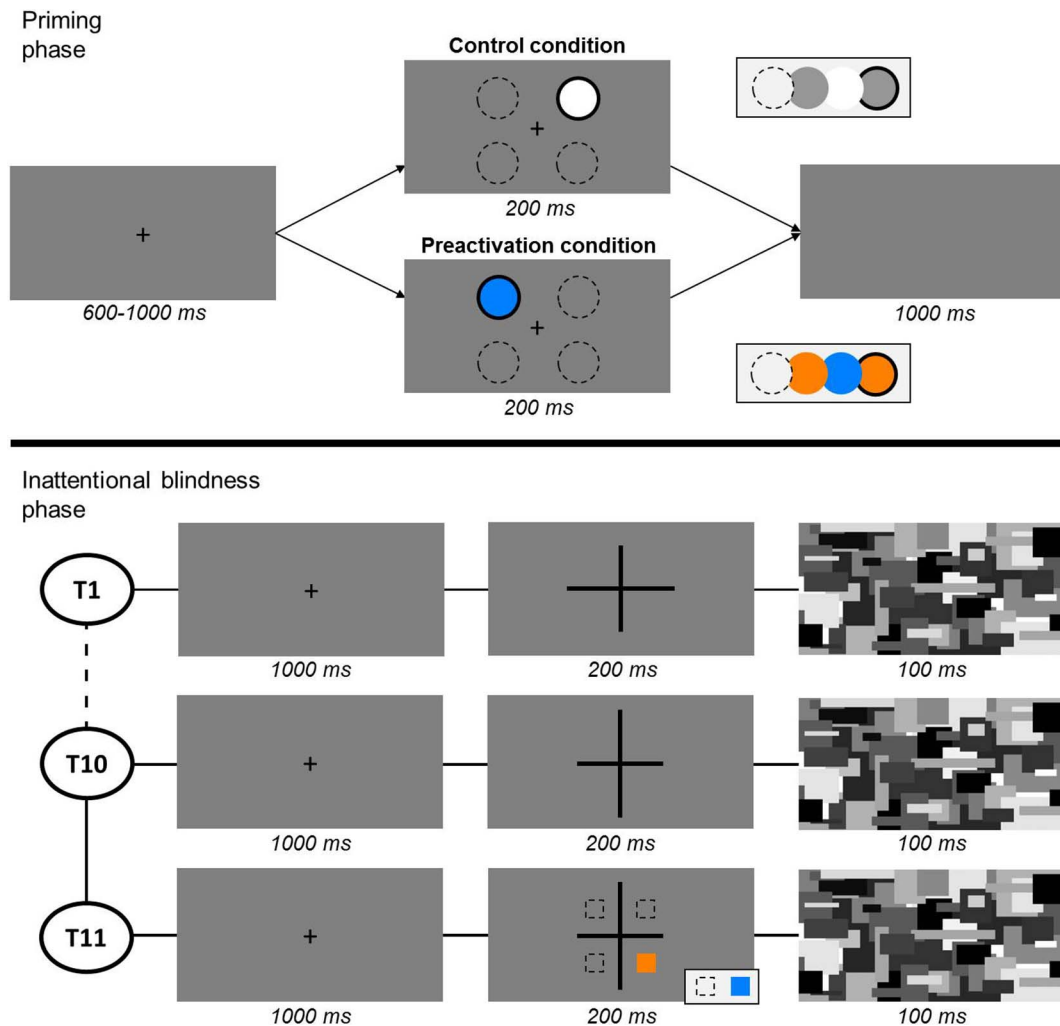


Fig. 1. The two phases of Experiment 1. Priming phase: One exemplary trial of each condition is depicted. Participants were instructed to count the circles that were framed with a black annulus. In the control condition, half of the stimuli were grey while half were white. In the preactivation condition, half of the stimuli were blue and half were orange. Circles were presented in one of four possible positions (dashed circles). Inattentional blindness phase: Participants judged whether the vertical or horizontal arm of a briefly presented cross was longer. They completed five easier (T1-T5; arms 6° and 4.5°) and five more difficult (T6-T10; arms 6° and 5.62°) trials. On the eleventh trial (T11; critical trial) a blue or orange square (0.9° × 0.9°) was presented in one of four possible positions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

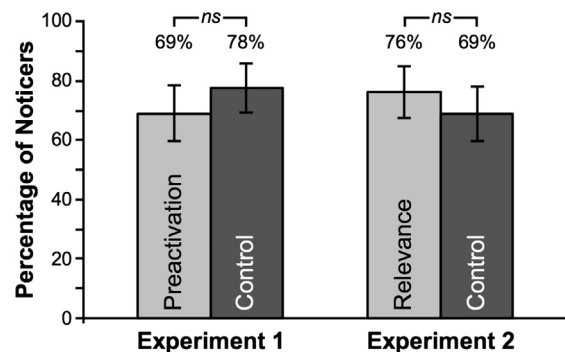


Fig. 2. Main results of the two experiments. The percentage of participants who noticed the unexpected object is shown, with error bars indicating the 95% confidence interval.

problems or a poor contrast. All statistical analyses conducted were two-tailed. As a standardized measure of effect size, risk ratios (RR) with 95% confidence intervals in square brackets are reported.

The results of Experiment 1 are depicted in Fig. 2. Overall, participants performed very well in the priming phase (preactivation condition: $M = 0.54$ errors; $SD = 0.89$ errors; control condition: $M = 0.60$ errors; $SD = 1.15$ errors). These data strongly suggest that the relevant stimulus dimension (color) was adequately preactivated. Nevertheless, noticing rates of the unexpected object for the critical trial did not differ between the preactivation and the control condition, $\chi^2(1) = 1.51$, $p = .22$, RR (preactivation/control) = 0.90 [0.76, 1.06]. Thus, individuals did not notice the unexpected object significantly more often when its color was perceptually preactivated and there was not even a tendency in the hypothesized direction.⁴ There was no significant difference in noticing rates of orange and blue unexpected objects across conditions ($\chi^2(1) = 0.99$, $p = .32$, RR (orange/blue) = 1.09 [0.92, 1.29]) and there was no difference in noticing rates between the preactivation and control condition for either object color. Also, the pattern remained the same when the data were analyzed separately for male and female participants. To sum up, the results of Experiment 1 did not indicate that repeated preactivation of a perceptual feature leads to a higher probability of conscious perception of a feature-related visual event.

3. Experiment 2

In Experiment 1, we tested whether simply preactivating a certain perceptual feature (a color) affects whether or not an unexpected object possessing this feature was subsequently detected more frequently than objects not containing it. Note that the underlying idea was that a neural activation representing a certain feature in a prior situation might persist in a later situation, where it might amplify related sensory events and allow them to pass the threshold of conscious awareness. Although we did not find support for this theorizing in Experiment 1, the general idea of potentially influential, persistent activation calls for a closer look at what might actually make a mental representation both strong and long-lasting, thus maximizing its potential influence in subsequent situations. It has been suggested that the strength and duration of any type of activation largely depends on its *motivational relevance* (Eitam & Higgins, 2010). This proposal provides a theoretical background for a series of findings on inattention blindness: Stimuli that are threatening, animated, or related to the observer's identity and current goals are less likely to go unnoticed than stimuli that do not possess these qualities (Calvillo & Jackson, 2014; Mack, Pappas, Silverman, & Gay, 2002; Most et al., 2005; New & German, 2015). The subjective value of the unexpected stimulus appears to affect its detectability, with highly relevant objects being missed less frequently.

Crucially, the framework provided by Eitam and Higgins (2010) not only posits that the strength and duration of a current activation, but also of a *preactivation* depends on its motivational relevance. Thus, when investigating the influence of preactivation on subsequent inattention blindness, one should factor in the role of motivational relevance. If a certain feature (e.g., a color) is not only activated in a previous situation, but also motivationally relevant to the observer, this preactivation should be stronger and longer lasting (Eitam & Higgins, 2010). Consequently, when an unexpected stimulus containing this feature appears in a subsequent situation, it should have a higher chance of being detected (see Kreitz, Schnuerch, Furley et al., 2015).

In Experiment 2, we therefore asked participants to first perform a simple detection task in which a certain color was of high motivational relevance. In a subsequent inattention blindness scenario, we assessed whether stimuli of the previously relevant color were noticed more frequently than stimuli of other colors. Previous work shows that stimulus features become motivationally relevant if they are necessary to complete a given assignment (e.g., a color that separates targets from distractors in a detection task). This means that task relevance is a valid proxy for motivational relevance (see, e.g., Bloom, Potts, Evans, & Drobles, 2013; Potts, 2004). In Experiment 2, we therefore manipulated the task relevance of certain colors by making them the to-be-identified feature in the first task. We hypothesized that the unexpected objects in the second task would be detected more frequently (i.e., there would be less inattention blindness) if the object's defining feature, its color, was motivationally relevant in the first task than if it was not.

3.1. Method

3.1.1. Participants

A total of 200 people who had not participated in Experiment 1 took part in Experiment 2. They gave written informed consent and received candy as compensation. All participants reported normal or corrected-to-normal vision. We excluded participants from analysis if they (a) indicated in the follow-up questionnaire that they had anticipated the unexpected object or knew that inattention blindness was the subject of the study (7 participants), (b) did not notice the unexpected object when they did not have to perform the primary task (full-attention trial; 5 participants), or (c) did not perform adequately on the task during the priming phase (the number of errors was more than 2 SD above the sample mean for 3 participants, which were subsequently excluded from further analysis). One additional participant was dropped from further analysis due to an error in data recording. Data from the remaining 184 participants were analyzed ($M = 22.5$ years; $SD = 5.5$ years; 54.6% female).

3.1.2. Materials and procedure

Except for the described differences, all materials and procedures were identical to those of Experiment 1. Participants were randomly assigned to one of the two experimental conditions: The color of the unexpected object in the inattention blindness task

⁴ The raw data along with a description of variables and coding can be accessed via <https://osf.io/fqu4d/>.

was either motivationally relevant during the priming phase (relevance condition) or was motivationally irrelevant (control condition). Motivational relevance was established by making the circles' color task-relevant. Stimuli during the priming phase were identical to those in Experiment 1 except for their color: In both conditions (relevance and control), half of the circles were orange (RGB: 255,127,0) and half were blue (RGB: 0,127,255). Again, half of the circles were framed by a black annulus, but this feature was task-irrelevant in Experiment 2. Instead, participants were instructed to either count the orange or the blue circles. There were 5–15 targets in each of the eleven blocks with no target count being used twice. Participants silently counted the targets for each block and entered the number after each block. Whether the target color was blue or orange was counterbalanced across participants.

3.2. Results and discussion

The results of Experiment 2 are depicted in Fig. 2. As in Experiment 1, participants performed very well in the priming phase (relevance condition: $M = 0.44$ errors; $SD = 0.79$ errors; control condition: $M = 0.43$ errors; $SD = 0.70$ errors). Even though this suggests that motivational relevance was adequately established, noticing rates of the unexpected object in the critical trial did not differ between the relevance and the control condition, $\chi^2(1) = 0.83$, $p = .36$, $RR(\text{relevance/control}) = 1.08$ [0.91, 1.28]. Thus, individuals did not notice the unexpected object more often when its color had been motivationally relevant in a previous task. There was only a slight tendency in the hypothesized direction. Even though orange unexpected objects were noticed significantly more often than blue objects across conditions [$\chi^2(1) = 4.17$, $p = .04$, $RR(\text{orange/blue}) = 1.19$ [1.00, 1.42]], there was no difference in noticing rates between the two conditions for either object color. Also, the pattern remained the same when the data were analyzed separately for male and female participants.

On the basis of Experiment 2, we cannot conclude that an unexpected objects is more likely to be detected if the object's defining feature was recently motivationally relevant. Participants were not less likely to experience inattentional blindness to a colored object when its color had been a selection criterion immediately before.

4. General discussion

In the present study, we tested whether targeted attributes of prior visual search remain prioritized for visual attention and awareness. More specifically, we investigated whether an unexpected object was less likely to be overlooked due to inattention if its defining perceptual characteristic was preactivated. In Experiment 1, we found no evidence that mere perceptual exposure to a color reduced the likelihood of detecting an unexpected stimulus of the same color in a subsequent situation. Even making a color motivationally relevant in a detection task in Experiment 2 did not affect whether an unexpected stimulus of the same color was noticed in a subsequent situation. Thus, neither a simple perceptual nor a more elaborate, attentional preactivation of the key features of an unexpected object affected whether or not it went unnoticed.

Since our findings are null effects, one should carefully consider whether our analyses were powerful enough to detect any relevant effects in the first place. To evaluate this, we performed a sensitivity analysis in G*Power (Version 3.1.9.2; Faul, Erdfelder, Lang, & Buchner, 2007) and tested how small an effect we were able to detect with sufficient power ($1 - \beta > 0.90$). This was done taking into account the final sample sizes after exclusion of individual participants (Experiment 1: $N = 188$; Experiment 2: $N = 184$). The analysis revealed that we were able to detect with a probability of .9 effects that were small to medium-sized ($w \geq 0.24$). For medium-sized to large effects ($w \geq 0.40$), we even achieved a power ($1 - \beta$) > 0.99 . Thus, our null effects are not easily attributed to a lack of statistical power and can be taken as evidence in regard to our research question.

In fact, the dissemination of such null findings might be seen as particularly relevant, because psychology and psychiatry are biased more strongly than other scientific disciplines toward reporting positive effects (Fanelli, 2010). Unfortunately, this creates a distorted, misleading understanding of the real facts in these fields, as documented by an alarmingly low reproducibility of major findings (Open Science Collaboration, 2015). Null-effect reports—including, but not limited to, unsuccessful replications (Hendrick, 1990)—can help establish a more balanced view of the state of affairs in psychological science.

4.1. Preactivation and inattentional blindness

Our findings replicate and broaden previous findings: We recently found no evidence of an effect of prior *semantic* activation on subsequent inattentional blindness (Kreitz, Schnuerch, Furley, et al., 2015; see also Légal et al., 2017). In the present study, we took a step back and asked an even simpler question: Does mere *perceptual* preactivation help a stimulus cross the threshold of awareness? Prior studies from adjacent fields suggest that perceptual processes, including identification, benefit from immediate preactivation (Goolsby & Suzuki, 2001; Tulving & Schacter, 1990; Wiggs & Martin, 1998). This does, however, not appear to extend to the issue of whether or not an object is detected despite inattention. Our data indicate that the conscious detection of unexpected sensory events does not benefit from sustained prior exposure to a related stimulation. This might be explained on the basis of contemporary models of consciousness. It has been proposed that sufficiently strong sensory stimulation that is not boosted by top-down attentional amplification merely constitutes “preconscious processing” (Dehaene et al., 2006). This form of representation is characterized by localized neural representations that are not widely (or “globally”) accessible in different areas and circuits of the brain (Dehaene et al., 2006). Perceptual preactivation, such as in Experiment 1, might lead to an accumulated, and thus more intense perceptual representation (see Tulving & Schacter, 1990). Nevertheless, the state of inattention, induced by the primary task distracting participants from the onset of the unexpected event, should still prevent a top-down amplification of this sensory input, however intense it might be. As a consequence, this input could still fail to reach awareness (see Dehaene et al., 2006).

Of course, one might argue that repeated exposure might simply leave no trace at all. That is, seeing a certain color over and over again might not change *anything* about the way a stimulus of this color is subsequently processed. According to this idea, not even the “preconscious”, localized sensory representation of the stimulus (see Dehaene et al., 2006) might have been amplified. Experience from the priming phase might simply trail off quickly, such that there is no chance that it might exert an influence in a subsequent situation. In fact, it has recently been argued that the quality and effectiveness of a representation largely depends on its relevance (Eitam & Higgins, 2010). In Experiment 2, we therefore made perceptual information motivationally relevant, as they now were the defining features that distinguished the to-be-detected targets from irrelevant distractors in the priming phase. This did, however, not affect the likelihood of inattention blindness, either. In this case, the preactivation created in the priming phase should clearly be strong and lasting (see Eitam & Higgins, 2010; see also Kreitz, Schnuerch, Furley, et al., 2015). The lack of an effect on subsequent detection therefore strongly supports the view gleaned from Experiment 1 that preactivation does not determine whether or not a certain object is subsequently consciously perceived despite inattention.

The fact that even prior motivational relevance did not affect inattention blindness is particularly noteworthy as the motivational relevance of a certain color was established by making this color the central selection criterion. In other words, this color previously defined the attentional set. It is well-established that the *current* attentional set (i.e., which criterion is used to direct one’s selective attention) strongly affects whether or not an object is missed or detected in inattention blindness scenarios: Objects associated with, or similar to, the current attentional set are noticed more frequently than others (see, e.g., Most et al., 2001). Experiment 2 of the present study shows that a *previous* attentional set has no effect whatsoever. There are two possible explanations for this finding. (a) Persistence over time: Attentional amplification based on a selection criterion might be tied to the moment, working only for one’s currently activated set. That is, whatever potentiation was produced in the first task might wear off quickly and, thus, fail to affect conscious noticing of an unexpected object in the second phase. Future research could make an effort to further shorten the time between the priming phase and the appearance of the unexpected object. Nevertheless, based on our data, we can already conclude that an attentional set does not affect conscious noticing of an unexpected object approximately 1–2 min later. (b) Independency of context: Attentional amplification based on a selection criterion might not be sufficiently context-independent to survive throughout the discontinuities of switching tasks. That is, the effectuality of a selection criterion might be bound to the specific context in which it originates. In the present study, the context changed after the selection criterion was established, thus possibly preventing a continuous attentional amplification. However, this explanation is at odds with findings reported by Most, Kuveldina, Dobson, and Kennedy (2011), who did not find an effect of color priming on inattention blindness, either. This was the case even though the unexpected object appeared in the same experimental context as the primes. Notably, though, they did find an effect of *negative* priming on inattention blindness: Unexpected objects of a color that was previously actively ignored were missed significantly more often than unexpected objects of a neutral or previously attended color. Based on these findings, future research might systematically focus on effects of negative priming in the context of inattention blindness.

The particular significance of inattention blindness lies in its ecological validity. Other experimental procedures, such as masking or the attentional blink, have the clear downside of being quite laborious and artificial (Schnuerch, Kreitz, Gibbons, & Memmert, 2016). Inattention blindness, however, clearly captures a realistic scenario that conceivably occurs regularly in real-life situations (see, e.g., Most & Astur, 2007). It therefore seems particularly expedient to infer what additional knowledge on its principles might mean for everyday instances of such failures of awareness. Previous studies suggest that inattention blindness might be a serious problem in traffic and medical diagnostics (Drew et al., 2013; Haines, 1991; Most & Astur, 2007). The present findings, along with our previous experiments (Kreitz, Schnuerch, Furley, et al., 2015), indicate that merely activating certain events that could be overlooked does *not* suffice to prevent such oversight. Neither prior exposure, nor a previous attentional set appear to avert subsequent inattention-based failures of awareness. This complements recent findings showing that a rough, general knowledge of potential misses does not decrease the proneness to this type of error (Beanland & Pammer, 2010; see also Kreitz, Furley, et al., 2015; Ward & Scholl, 2015). Thus, the need for immediacy, timeliness, and specificity of counter-measures intended to thwart inattention blindness should be considered by practitioners in the respective fields. For example, to increase the likelihood to detect what might otherwise go unnoticed, radiologists and pilots should maintain a currently activated attentional set (e.g., of potential tissue anomalies or obstacles on the runway) instead of merely relying on previous engagement with these issues or on one’s general knowledge that inattention blindness might occur. Specifically, if an air traffic controller should be able to detect rare, but immensely important yellow objects on his or her screen, an additional continuous task should incorporate yellow objects, too. At best, yellow objects have to be attended (e.g., other displays and signals that demand more frequent responses and, thus, create a continuous attentional set) while other colors have to be ignored.

4.2. Limitations of the present study

The present study has some limitations and shortcomings that should be addressed here and considered in future investigations. As in most types of research, our findings were based on a somewhat specific constellation of settings and manipulations that were selected to investigate the question at hand. Potentially, our findings that neither perceptual preactivation nor prior motivational relevance affect the likelihood of inattention blindness are limited to certain features of the unexpected object, such as its color. Also, the influence of moderating variables, such as preactivation, might be different for different inattention blindness paradigms. We cannot rule out the possibility that paradigms in which the unexpected object is presented for longer durations (e.g., for several seconds: Most, Simons, Scholl, & Chabris, 2000; Simons & Chabris, 1999) are more sensitive to the effects of preactivation. Thus, our findings might not generalize across all inattention blindness paradigms and all preactivated concepts or features. Interestingly, however, the effect of the current attentional set has been demonstrated reliably across different inattention blindness tasks

(Koivisto & Revonsuo, 2007; Most et al., 2001) as well as different features of the unexpected object (Most et al., 2005), including color (Most & Astur, 2007). Thus, at the very least, we can conclude on the basis of our present data that preactivating a specific feature—even by making it motivationally relevant for a certain time—does not have the same effect as a current attentional set involving this very feature.

As color perception declines in the periphery, it might be argued that the color of the unexpected object (which was presented 2° from fixation) was not sufficiently processed to allow preactivation of this very color to affect its detection. Although this is an important issue to consider, several arguments speak against the hypothesis that our findings can be explained by the spatial position of the unexpected object: (a) Decline of color perception is more dramatic for red and green while we specifically used blue and orange for the unexpected object (Hansen, Pracejus, & Gegenfurtner, 2009), (b) processing of color is problematically restricted only at approximately 25–30° and absent at approximately 40° from fixation (Abramov, Gordon, & Chan, 1991; Hansen et al., 2009; Johnson, 1986), and (c) the unexpected object was presented within the spatial focus of attention (between the arms of the cross that was attended as part of the primary task) which should have enhanced its neuronal processing (e.g., Kiefer et al., 2011; Posner & Driver, 1992). Nevertheless, future studies might want to conceptually replicate the present findings by preactivating a different feature of the unexpected object.

In addition, we chose to operationalize motivational relevance as task relevance in the priming phase. Even though previous work indicates that stimulus features become motivationally relevant if they allow us to successfully complete a given assignment (Bloom et al., 2013; Potts, 2004), our implementation of task relevance might not have sufficed to induce a preactivation strong enough to modulate noticing rates of the unexpected object. Presumably, such preactivation might have to be stronger and/or more longlasting. A higher degree of preactivation might be established by other forms of motivational relevance, such as monetary value (e.g., Anderson, Laurent, & Yantis, 2011), self-relatedness (Mack et al., 2002), or biological relevance (Ellis, 2001). A more longlasting preactivation might be equivalent to what is also called “inherent relevance”. Previous research on inattention blindness shows that inherent relevance, such as one’s own name (Mack et al., 2002), animated objects (Calvillo & Jackson, 2014), or faces (Devue, Laloyaux, Feyers, Theeuwes, & Brédart, 2009; Mack & Rock, 1998), can indeed reduce susceptibility to inattention blindness. The relevance of faces and animated objects might be biologically pre-defined but the relevance of one’s own name has to be learned over life. It is an open and interesting issue how extensive such learning has to be to generate a preactivation that suffices to modulate the threshold of awareness in an inattention blindness setting. Future research might utilize overlearning of stimulus-reward associations to investigate this question.

Finally, one might argue that the detection rates in the present experiments were poorly suited to demonstrate effects of preactivation. Ideally, they would be around 50% in the control condition to achieve the highest possible sensitivity to experimentally-induced shifts in either direction (i.e., lower or higher detection rates). However, detection rates in the present study were between 69 and 78%, which makes it unlikely that effects could not have been detected in the present study due to ceiling effects. This rate, although higher than it ideally would have been, still left enough room for potential experimental effects to surface.

4.3. Conclusion

Inattention blindness is an ecologically valid failure of awareness. Understanding what affects whether or not inattention blindness occurs is of high theoretical and practical relevance. Adding to recent findings suggesting that semantic preactivation does not affect inattention blindness, the present data clearly indicate that neither perceptual preactivation nor prior motivational relevance influence the likelihood of missing an unexpected object when attention is otherwise engaged. Future investigations should explore the limits of these findings; for example, by testing whether different types of prior motivational relevance, such as monetary reward value, do affect subsequent inattention blindness.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.concog.2018.01.003>.

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